

Influence of Simulated Aerodynamic Forces on Weight Perception and Realism in Virtual Reality

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ABSTRACT

In this research, the impact of aerodynamics on the perception of weight in virtual environments was studied. A real-time aerodynamic simulation was created for objects with different shapes and weights, and an experiment was conducted to determine the correlation between subjects' weight perception and the presence of simulated aerodynamic forces on virtual objects. The results showed that the weight of light objects was more accurately conveyed and the perceived realism was improved when simulated aerodynamic forces were included.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices

1 INTRODUCTION

Virtual training environments offer the opportunity to learn skills that would otherwise be complex, expensive, or dangerous to learn in reality. Immersion is a key factor in the success of training in these environments, and can be improved by recreating a realistic environment. To increase immersion, it is desirable for users to be able to feel the weight of virtual objects. Visual cues can influence the perception of weight in virtual environments. Various methods have been explored to manipulate weight perception using these cues. In the real world, personal experience with the flight characteristics of different objects can also serve as a guide for weight estimation. However, in a simple physics simulation, objects with different weights may move in the same way if factors such as air resistance are not taken into account. Our aim in this study is to investigate whether simulating aerodynamic forces in a virtual environment can improve the perception of weight and reality.

2 RELATED WORK

Weight perception is known to be influenced by various factors such as somatosensory information, size [10], shape [5], material [2], and mass distribution [1]. Haptic interfaces, which use hardware to apply forces to the user's limbs, have been used to create realistic weight sensations. However, these methods may be expensive and may restrict user movements. Pseudo-haptic interfaces, on the other hand, rely on software to create visual stimuli and are cheaper and more flexible, but may not be as precise [6]. Some approaches for creating the sensation of different weights involve manipulating the control-display ratio [8, 11] or physical properties of virtual objects [7] that are grasped or held by the user. There is relatively little research on the weight perception of moving objects that have no contact with the user. Zintus-art et al [12] studied the weight sensation of virtual objects sliding down an inclined plane on a tablet computer,

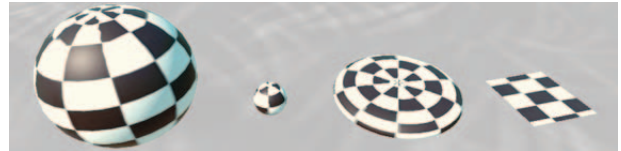


Figure 1: The four object shapes used in the study. From left to right the big ball, the small ball, the frisbee and a postcard.

while our research investigates the influence of motion in a more immersive environment with a head-mounted display.

3 EXPERIMENT

We are studying the potential influence of simulated aerodynamic forces on weight perception and perceived realism of virtual objects in virtual reality.

3.1 Simulation Environment

A virtual simulation environment was created using the *Unity Game Engine* (version 2020.3.30f1) for the purpose of studying the impact of aerodynamic forces on the perceived weight of thrown virtual objects. There are four throwable objects each with three weights (Fig. 1): a big ball (diameter 0.24 m, weights $5 \cdot 10^6 \text{ kg}$), a small ball (diameter 0.065 m, weights $0.5 \cdot 10^6 \text{ kg}$), a frisbee (diameter 0.275 m, weights $1.75 \cdot 10^6 \text{ kg}$), and a postcard ($0.148 \text{ m} \times 0.148 \text{ m} \times 0.0001778 \text{ m}$, weights $0.7 \cdot 10^6 \text{ kg}$), $e \in \{-10, -1, 0\}$).

Simulated forces for various throwable objects in a virtual reality game include gravity, lift, drag, and torques. Coefficients and formulas for these forces are based on Hummel et al. [4] and Santos et al. [3]. The flight characteristics of a postcard are recreated by adjusting the weighting of the formulas developed for frisbees. The forces are scaled to work within *Unity's* physics engine, which is based on *Nvidia PhysX*. The engine's results are not deterministic, but mostly plausible. The simulation runs at 72 frames per second, and velocity vectors are smoothed over the last 14 frames and scaled by a factor of 1.2 for more consistent throwing behavior. The *Meta Quest 2* virtual reality headset and controllers are used in the experiment, and participants' interpupillary distances are measured.

The *Meta Quest 2 Touch Controllers* are used to interact with virtual objects in a virtual reality environment. Raycasting is used to select objects and the grab button is used to grab and throw objects that have been assigned grab offsets for realistic throwing motion. The trigger button is used to interact with graphical user interfaces. The controllers may not accurately represent the shape, form, or center of mass of the virtual objects, which can affect the realism of the experience.

3.2 Experiment Design and Test Session

The study consists of a virtual reality component and a questionnaire. The virtual reality component contains two sections. In the first section, participants are presented with pairs of objects, one with and one without simulated aerodynamic forces, and are asked to select which felt heavier. The objects and order of presentation

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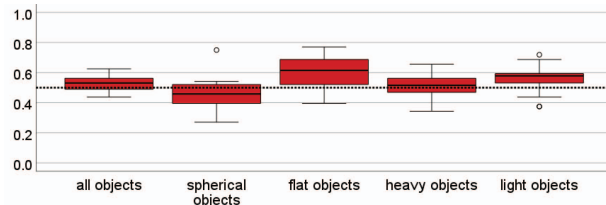


Figure 2: Percent of times objects with aerodynamics were marked as lighter per participant.

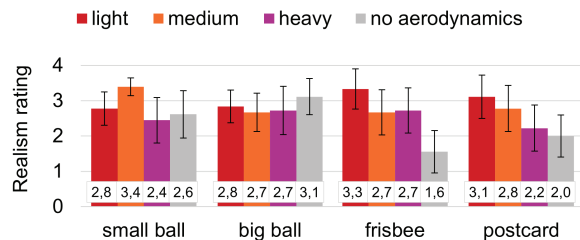


Figure 3: Realism evaluation (0 = unrealistic, 1 = somewhat unrealistic, 2 = neither realistic nor unrealistic, 3 = somewhat realistic, 4 = realistic)

are randomly generated. In the second section, participants throw each object as many times as they like and rate the realism of the flight behavior. After the virtual reality component, participants are asked to fill out a questionnaire with demographic and other relevant questions, as well as the igroup presence questionnaire (IPQ) [9].

25 participants were recruited for the study, with 18 (8 female, 10 male) remaining after technical issues were accounted for. The ages of the participants ranged from 18 to 65 years old, with a median age of 25 years. The study was partially conducted in a home setting and participants were not informed about which objects had simulated aerodynamic forces enabled. Participants were given a short introduction to the controls and were asked to throw with their dominant hand. One participant reported light headaches after the study.

4 RESULTS

For each participant, the percent of times objects with aerodynamic forces were perceived as lighter (f) was calculated Figure 2. Non-parametric tests (Kolmogorov-Smirnov and Wilcoxon signed-rank) were used to test the null hypotheses. The data was also analyzed by subgroups, but no significant results were found.

H1: Simulated aerodynamic forces do not significantly affect the perception of weight of virtual objects in virtual reality. f_{all} (median of 0.53) was tested against the hypothetical median of 0.50 with $Z = 1.968$ and $p = 0.049$. We have to reject H1.

H2: The shape of virtual objects with simulated aerodynamic forces does not significantly affect the perception of their weight. $f_{spherical}$ with a median of 0.46 was tested against f_{flat} with a median of 0.61. Results show $Z = 2.443$ and $p = 0.015$. We have to reject H2.

H3: The weight of virtual objects with simulated aerodynamic forces does not significantly affect the realism of their flight behavior. Figure 3 shows the average realism rating for objects with and without simulated aerodynamic forces. For frisbees ($Z = -2.70$ and $p = 0.007$) and postcards ($Z = -2.174$ and $p = 0.030$) significant results were found. For spherical objects, no significant improvement was found with simulated aerodynamic forces. We cannot reject H3.

To evaluate the IPQ we investigated each of the subscales and the overall scale against a hypothetical median of 3 with a Wilcoxon

signed-rank test. The overall presence scale ($W = 5952$, $p < 0.01$) and the spatial presence ($W = 210.5$, $p < 0.01$) show a statistically significant difference. Thus for our environment overall presence and spatial presence is significantly higher than the neutral IPQ value of 3.

5 DISCUSSION

Simulated aerodynamic forces significantly impact the perception of weight of virtual objects in virtual reality. Spherical objects are perceived as heavier with simulated aerodynamic forces, while flat objects are perceived as lighter. Light objects with simulated aerodynamic forces are also perceived as lighter. The flight behavior of normal weight frisbees and postcards is perceived as significantly more realistic with simulated aerodynamic forces, but this is not the case for spherical objects.

6 CONCLUSION AND OUTLOOK

Simulated aerodynamic forces should be included in virtual reality applications that value realism and include thrown objects. It would be interesting to investigate the influence of simulated aerodynamic forces on objects of different shapes and weight, including shapes such as cubes or tori.

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